

#HW 1

● Graded

Student

Samantha Rangel

Total Points

70 / 100 pts

Question 1

Q1

18 / 20 pts

- 2 pts There is an error in your calculation. Feet should be converted to inches. ft^3 can be converted to inch^3 by multiplying it with 12^3.
- 0 pts Click here to replace this description.
- ✓ - 2 pts Since the Patm is given in lbf so you should get your gauge and system pressure in lbf too. So you should divide rho_{gh} by the gravity.
- 2 pts Absolute or system pressure can be obtained by adding up the gauge pressure with Patm.
- 2 pts The effective height should be 15*sin 40 degrees. You would have got a different answer if you had used 40 degree instead of 30 degree since the question mentioned a 40 degree inclination.
- 5 pts Late

Question 2

Q2

15 / 20 pts

- 0 pts Click here to replace this description.
- 4 pts Since, the specific volume has been given, in order to get the density, you need to take the reciprocal of the specific volume.
- 1 pt The answer you got is in Pascal. To convert it to kPa you must divide it by 1000.
- ✓ - 5 pts Incomplete submission
- 5 pts Late

Question 3

Q3

10 / 20 pts

- 0 pts Click here to replace this description.
- 2 pts In the question it has been mentioned that the pressure changes from 1 bar to 200 bar in increments and volume should be calculated next. You had increased volume and then calculated pressure. You are supposed to increase the pressure.
- 2 pts Pressure should be on the y axis and volume should be on the x axis
- 2 pts Please remember to share your data from the next time. The question also mentioned that you should share the screenshot of both the plot and data.
- 2 pts The question said to add 20 increments. So you should have reported 20 data sets.
- 2 pts Please remember to add titles on your axes from the next time.
- ✓ - 10 pts Incomplete
- 5 pts Late
- 1 pt Add units on axes.

Question 4

Q4

12 / 15 pts

– 0 pts [Click here to replace this description.](#)

✓ – 2 pts The temperature change is not the same for all. It is the same for C and K. It is the same for F and R. You should clearly state it.

– 2 pts The conversion scale for F to C is wrong. It is $(F-32)/1.8$ i.e. the entire $(F-32)$ gets multiplied by $5/9$ or divided by $9/5$.

– 5 pts Incomplete

✓ – 1 pt You should show the conversion of both 70 degree F and 42 degree F. You have not shown conversion for the 70 degree F.

– 5 pts Late

Question 5

Q5

15 / 25 pts

– 0 pts [Click here to replace this description.](#)

– 2 pts You are supposed to calculate the absolute pressure in 5a. So you add the P_{atm} too for getting that.

– 1 pt Pressure inside the submersible is same as the atmospheric pressure.

– 1 pt You should deduct P_{atm} from P_{abs} to get gauge pressure

✓ – 10 pts Incomplete

– 10 pts Late

Question assigned to the following page: [1](#)

#1

a)

$$\begin{aligned}
 P_{\text{absolute}} &= P_{\text{atm}} + (h \cdot \rho \cdot g) \\
 &= 14.7 + (9.642 \cdot 0.0314 \cdot 32.2) \\
 &= 24.45 \text{ lbf/in}^3
 \end{aligned}$$

$$\sin 40 = \frac{h}{15 \text{ in}}$$

$$\begin{aligned}
 h &= 15 \sin(40) \\
 &= 9.642
 \end{aligned}$$

$$\rho = \frac{54.2 \text{ lb}}{\text{ft}^3} = \frac{54.2 \text{ lb}}{(12 \text{ in})^3} = 0.0314 \text{ lb/in}^3$$

b)

$$\begin{aligned}
 P_{\text{gage}} &= P_{\text{absolute}} - P_{\text{atm}} \\
 &= 24.45 - 14.7 \\
 &= 9.75 \text{ lbf/in}^3
 \end{aligned}$$

c) liquids moves with less difficulty on a inclined plane compared to a U shaped plane.

#4

$$a. T = 42^\circ\text{F} = 5.56^\circ\text{C} = 278.56\text{K} = 501.408\text{R}$$

$$b. \Delta T = 70^\circ\text{F} - 42^\circ\text{F} = 28^\circ\text{F} = -2.22^\circ\text{C} = 270.78\text{K} = 487.404\text{R}$$

c. The Temperature change from $^\circ\text{F}$ to $^\circ\text{R}$ is 1.8 times difference

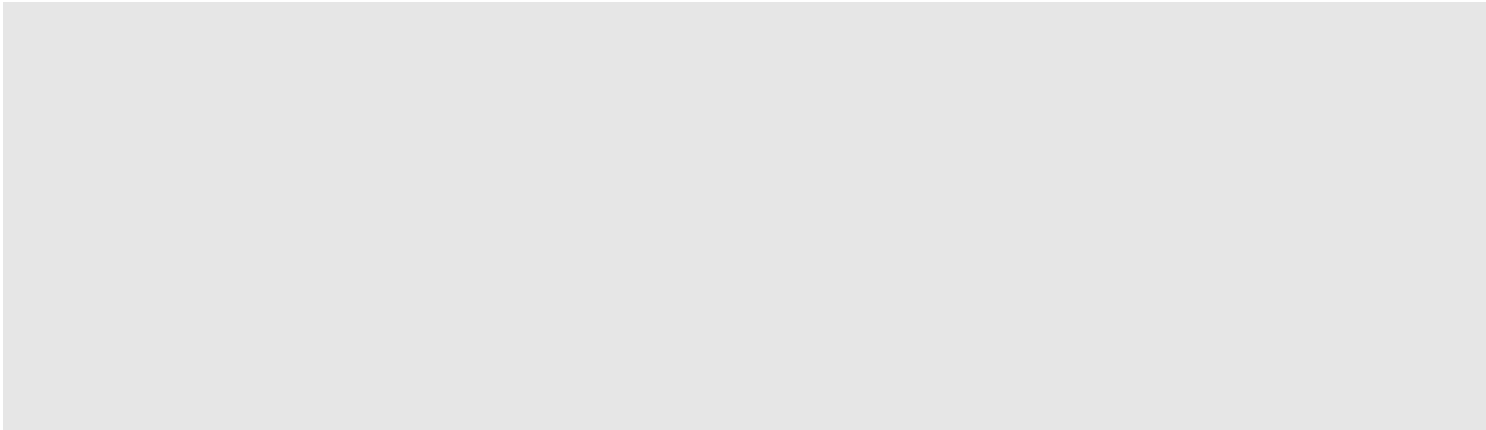
e. The temperature change from $^\circ\text{K}$ to $^\circ\text{C}$ is it goes from a larger to smaller scale.

#2

$$\rho = \frac{1}{V}$$

$$h = 0.12$$

$$\begin{aligned}
 P_{a1} - P_{a2} &= \rho \cdot g (H_1 - H_2) + P_{\text{atm}} \\
 &= \left(\frac{1}{0.00122} \right) (9.81) (12 - 0) + 101 \text{ kPa}
 \end{aligned}$$



Question assigned to the following page: [2](#)

#1

a)

$$\begin{aligned}
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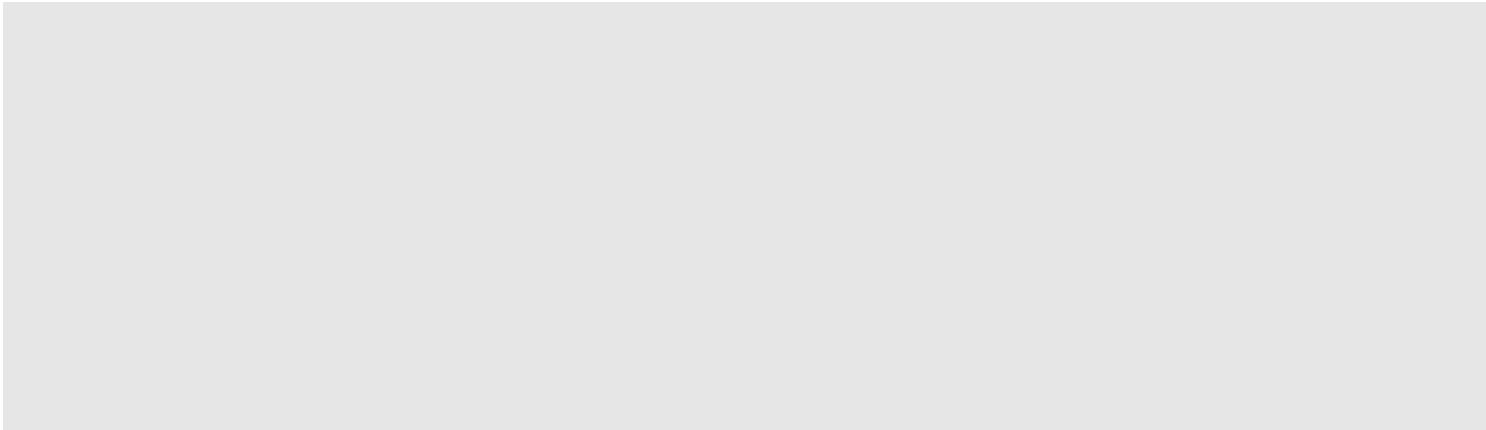
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Question assigned to the following page: [3](#)

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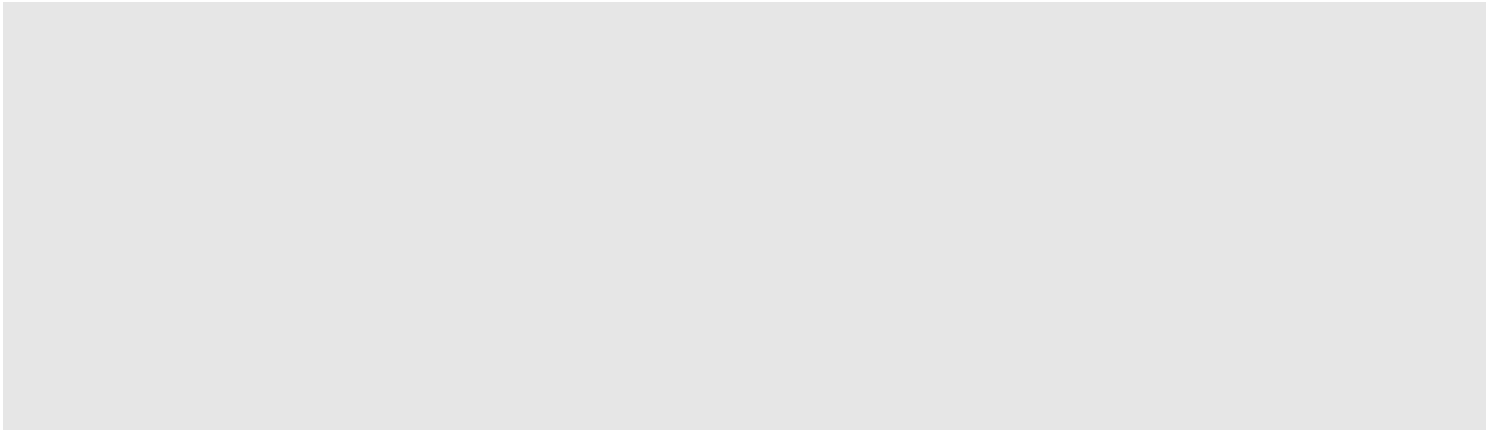
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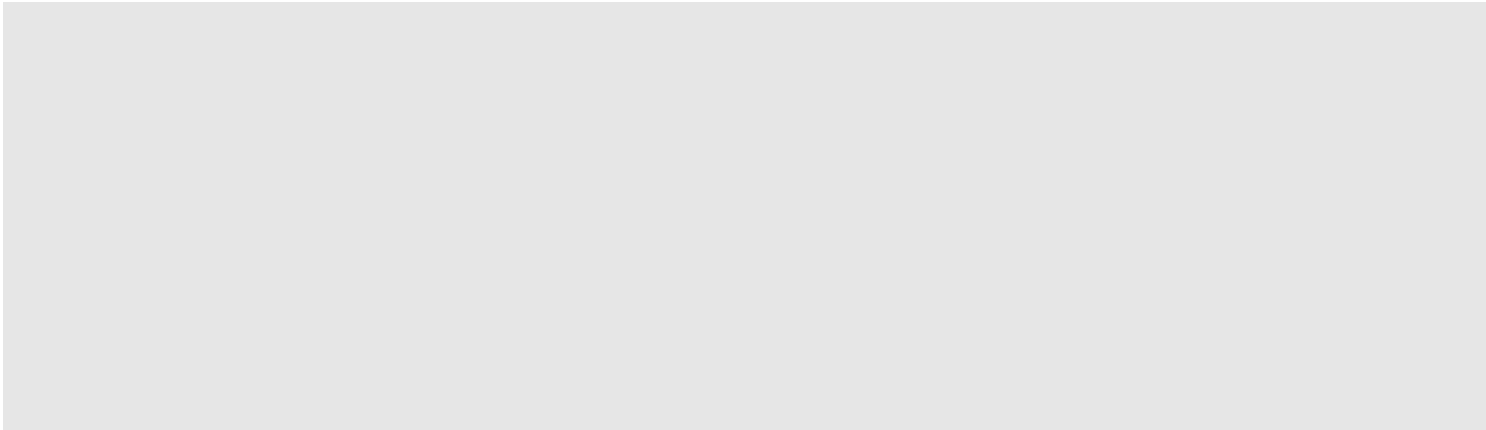
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